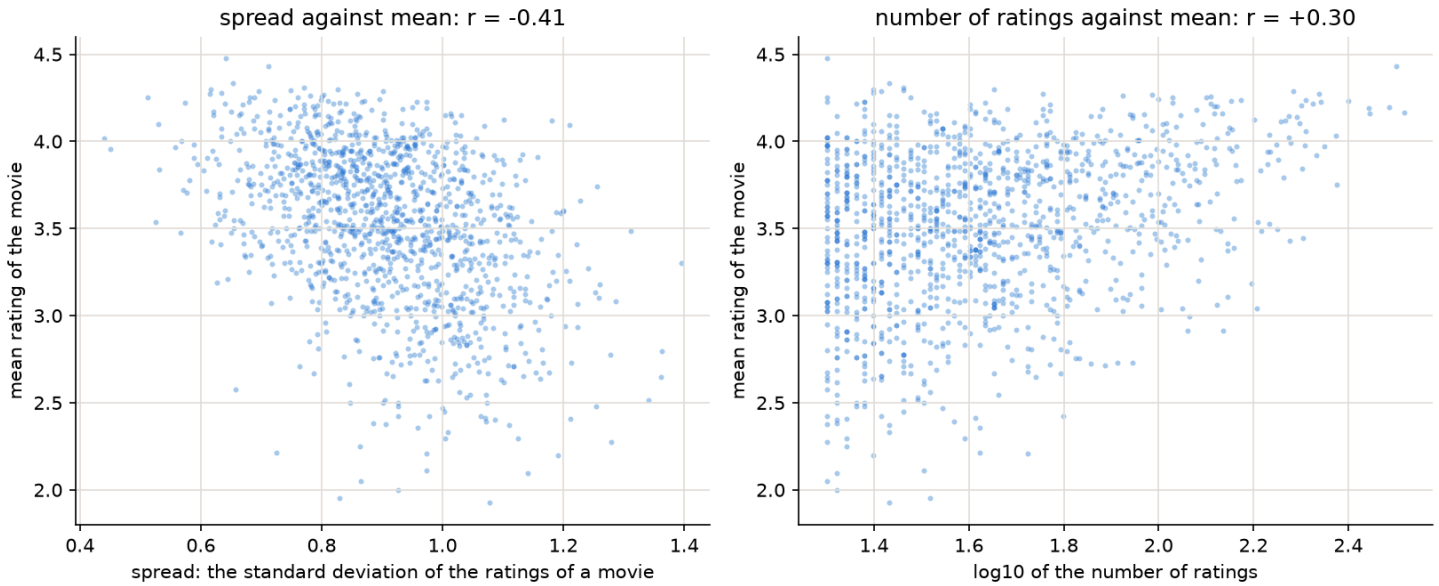
CS 301 · MEETING 14 · ACT 1

Correlation

Correlation measures the direction and strength of a linear association. A matrix product computes it for every pair of features.

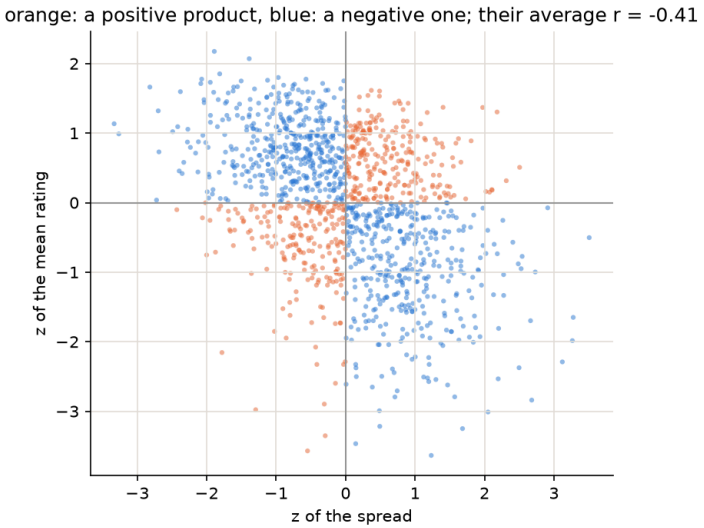
Scatter plots of movie features



- Each movie is one point: x from one column of the table, y from another
- Left: the spread of a movie's ratings against their mean
- Right: its number of ratings, on a log scale, against the mean

Standardized coordinates

$$z = \frac{x - \mu}{\sigma}$$



- Standardize each column: subtract its mean, divide by its standard deviation
- Above both means or below both: a positive product (orange); one of each: negative (blue)
- **r is the average product**, between −1 and 1

Correlation as a dot product

$$r = \frac{1}{n}\sum_{i=1}^{n} z_{x,i}\, z_{y,i} = \frac{z_x \cdot z_y}{n} = \cos\theta$$

```
def correlation(x, y):
    return standardize(x) @ standardize(y) / len(x)
```

```
r of spread and mean: -0.413 as a dot product, -0.413 by np.corrcoef
```

- A sum of products over n: a **dot product** divided by n
- θ: the angle between the two centred columns, seen as vectors with one entry per movie

Genre correlations

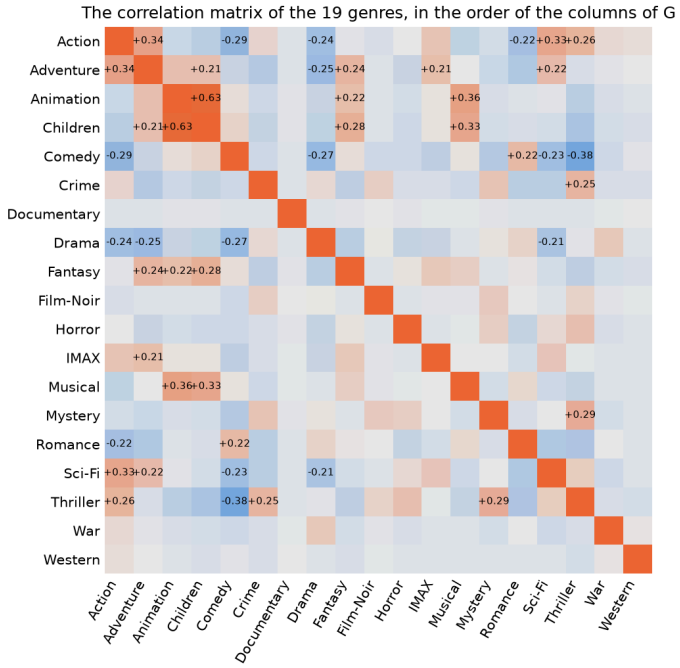
$$C = \frac{1}{n} Z^T Z, \qquad C_{ij} = \frac{1}{n} \sum_{k=1}^{n} Z_{ki} Z_{kj}$$

```
Z = (G - G.mean(axis=0)) / G.std(axis=0)   # every genre column standardized
C = Z.T @ Z / len(Z)                       # the correlation matrix, 19 by 19
assert np.allclose(C, np.corrcoef(G, rowvar=False))
```

```
Z is 1,297 by 19; Z.T @ Z / n is 19 by 19: 1,297 multiplications for each of its 361 cells
```

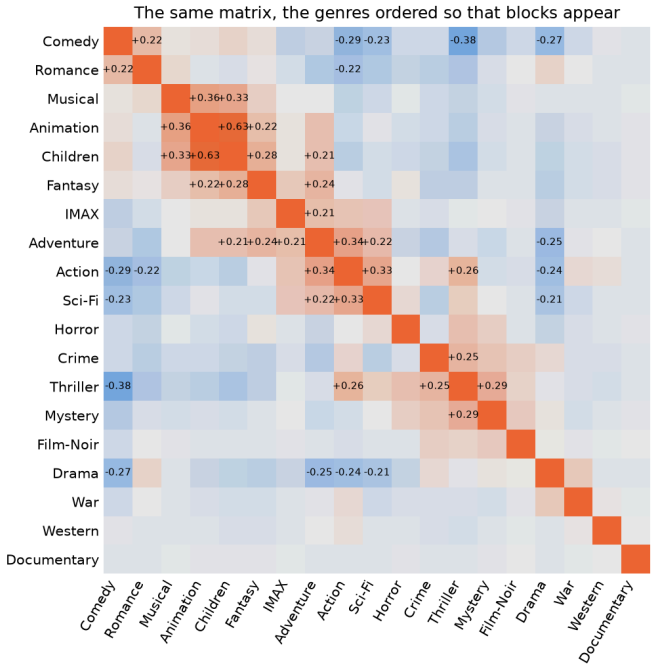
- Cell (i, j): the dot product of columns i and j, over n
- 171 distinct pairs of genres, excluding self-comparisons
- 1 on the diagonal; symmetric

The genre correlation matrix




```
the three most negative pairs: ['Comedy and Thriller: -0.38', 'Action and Comedy: -0.29', 'Comedy and
Drama: -0.27']
the three most positive pairs: ['Animation and Children: +0.63', 'Animation and Musical: +0.36',
'Action and Adventure: +0.34']
```

Reordered genre correlations



- Rows and columns move together; no number changes
- The order: hierarchical clustering using 1 − C as dissimilarities
- Positive blocks include Animation with Children, and Crime with Thriller
- Drama has negative correlations with Action and Comedy